Unit 7: Multimedia Authoring, Hypermedia and Interaction

Professional challenge: turn an assortment of photographs, recordings, documents and descriptions into a resource that a visitor can explore and understand.

This unit joins media production with information architecture, navigation, user-interface design, usability and accessibility.

BLISS case: interactive library orientation. BRIM case: a curated, permission-checked digital archival exhibition.

KEY TAKEAWAY • A multimedia resource succeeds when users can find, understand and use its content.

Learning outcomes and practical products

Explain multimedia authoring, hypertext, hypermedia, non-linear navigation and the components of an interactive information system.

Draw a page map, storyboard a user journey and distinguish appropriate authoring tools and publishing formats.

Create and test a small three-to-five-screen interactive guide with clear paths, keyboard access, alternative text and rights notices.

KEY TAKEAWAY • Design, implement, test and document a small usable prototype.

From media files to an authored experience

A folder of image, audio, video and text files is a collection of assets, not automatically a coherent multimedia experience.

Authoring gives assets structure: screens, layout, messages, navigation controls, sequencing, feedback and links.

For example, a visitor chooses a historical period, opens a photograph, reads its description and returns to the timeline.

KEY TAKEAWAY • Structure and user choices transform files into an information service.

What multimedia authoring means

Multimedia authoring is the planning, integration and arrangement of multiple media elements into a publishable learning or information resource.

An authoring environment may provide page layouts, timelines, templates, links, hotspots, actions, preview and export.

Authoring is different from editing a single photo or video: it defines how the elements work together for a user.

KEY TAKEAWAY • An authored system combines content, relationships, behaviour and presentation.

Hypertext and hypermedia

Hypertext connects textual information using links; a user follows references between documents or sections.

Hypermedia extends linked navigation across images, audio, video, maps, text and interactive objects.

Example: selecting a map region opens an archival photograph, an oral-history excerpt and contextual catalogue data.

KEY TAKEAWAY • Hypermedia is about linked resources and user-driven movement, not just animated pages.

Linear, non-linear and mixed structures

Linear: fixed order, such as a narrated training video or compulsory safety induction.

Non-linear: users choose among locations, themes or topics; a hypermedia exhibition has multiple possible paths.

Mixed: a guided introduction leads to freely browsable sections, with clear back and home controls throughout.

KEY TAKEAWAY • Choose structure according to learning goals and the freedom users require.

Nodes, links, anchors and destinations

A node is a page, screen, section or media item. A link connects a source node to a target node.

The anchor is the clickable or focusable label, image region or button that exposes the link; its text should communicate purpose.

A broken link points nowhere. An orphan page has no discoverable inbound route. Both frustrate access and quality control.

KEY TAKEAWAY • Draw navigation as a graph before building screens.

Information architecture and content grouping

Information architecture organises content into meaningful categories, labels, hierarchies and paths for finding information.

Begin with user questions: “Where are the opening hours?”, “How do I access an archival record?”, “What am I looking at?”

Group related items, use consistent terminology, and avoid making one page a dumping ground for every asset.

KEY TAKEAWAY • Good navigation starts with clear content organisation.

A practical sitemap: digital library guide

Home → Search the catalogue / Borrow resources / Research support / Contact the library.

Each branch explains a single task and contains a relevant example, a next step and a route back to Home.

A student should reach the catalogue demonstration in one or two intentional decisions, not through guesswork.

KEY TAKEAWAY • Write labels based on actual tasks, not vague words like “More” or “Click here”.

A practical sitemap: archival exhibition

Home → Time periods → Featured items → Item detail → Rights and source information.

At each item include a title, date/uncertainty, provenance, contextual description, permitted media and a return path.

Use only fictional or cleared material in class; archival access must follow permissions and sensitivity review.

KEY TAKEAWAY • An attractive exhibit must not erase context, provenance or restrictions.

Storyboards and screen-by-screen planning

A storyboard is a numbered plan of screens or scenes indicating what appears, what the user does and what happens next.

A useful screen row contains screen ID, goal, heading, media, interaction, next/back destination and accessibility notes.

The storyboard helps identify missing transitions and unnecessary media before software implementation.

KEY TAKEAWAY • Storyboard behaviour as well as appearance.

User journeys: start, choice, result, recovery

A user journey describes a specific goal from entry to successful completion, including likely detours and errors.

Example: a new student opens the library guide, chooses “Search”, follows the catalogue demonstration and finds the help option.

Provide recovery paths: Home, Back, clear labels, helpful empty states and no irreversible surprise actions.

KEY TAKEAWAY • Test whether a real student can complete the intended task.

Multimedia authoring tools: selecting a category

Presentation tools can build linear tutorials and hyperlink-based guides; visual website builders can publish simple structured pages.

HTML/CSS/JavaScript can provide flexible offline web interaction; learning-authoring tools can offer branching and quizzes.

Compare output portability, offline use, accessibility controls, cost, media handling and long-term maintenance—not just templates.

KEY TAKEAWAY • The best authoring tool fits the audience, output and institution.

Static web authoring is practical for this course

A static website can combine HTML structure, CSS styling, JavaScript interactions, images, video, captions and downloads.

A simple guide can work offline from a folder and later be published on GitHub Pages, without a database or user accounts.

Static does not mean non-interactive; however, local progress is not a formal submission, and restricted material should not be published.

KEY TAKEAWAY • A small static site is suitable for safe prototypes and general information.

The three roles of HTML, CSS and JavaScript

HTML gives semantic structure: headings, navigation, links, lists, forms, captions and media controls.

CSS controls appearance, spacing, contrast, responsive layout and visible focus states.

JavaScript manages states such as chosen exhibit item, question feedback or a branching flow; core information should remain accessible.

KEY TAKEAWAY • Separate meaning, presentation and interaction to simplify maintenance.

Buttons, links and hotspots: use them correctly

Use a link when the action navigates to another page or a named location. Use a button when an action changes the interface.

A hotspot should have a meaningful accessible name and a keyboard-operable equivalent; avoid pointer-only invisible areas.

Choose labels such as “View 1960s photographs” instead of “Click here”; identify downloads and external destinations.

KEY TAKEAWAY • Make actions predictable and operable without a mouse.

Feedback, controls and system status

A user should know which screen or item is selected, which action just occurred and what to do next.

Provide concise feedback: “2 of 4 questions correct”, “Item 3 of 5”, “No results for that term”, or “Download starts”.

Avoid hidden automatic playback, surprise navigation, inaccessible drag-only controls and misleading success messages.

KEY TAKEAWAY • Visibility and predictable feedback reduce user confusion.

Usability: usefulness, ease and efficiency

Usability asks whether specified users achieve specified goals effectively and efficiently, with satisfaction in context.

Assess task completion, navigation errors, unnecessary steps, comprehension, frustration and recovery—not beauty alone.

A quick five-task test may reveal serious problems even when the pages look professionally designed.

KEY TAKEAWAY • Observe tasks and outcomes instead of relying only on opinions.

Accessibility: why it matters

Users may rely on keyboards, screen readers, magnification, captions or colour-independent cues.

Design for clear headings, readable contrast, descriptive alternatives, keyboard operability and a logical focus sequence.

WCAG 2.2 provides testable criteria; a small checklist helps, but cannot establish full WCAG conformance.

KEY TAKEAWAY • Accessibility is part of authoring and testing, not decoration.

Semantic structure and readable navigation

Provide a descriptive page title, one primary heading, ordered subheadings, a labelled navigation region and meaningful link text.

Use real headings and lists rather than a long series of visually styled generic blocks.

Make current location identifiable through text or a clear selected state, not colour alone.

KEY TAKEAWAY • Semantic structure supports comprehension and assistive technology.

Keyboard operation, focus and target size

A keyboard user should move through interactive controls with Tab/Shift+Tab and activate expected actions with Enter or Space.

A visible focus indicator shows which control is active. Avoid trapping focus in a custom exhibit or menu.

Make touch targets usable on mobile; test both large and small screens, and try every essential interaction without a mouse.

KEY TAKEAWAY • Never make the only route depend on hovering or dragging.

Contrast, typography and responsive presentation

WCAG 2.2 AA normal-text contrast is generally at least 4.5:1; large text generally at least 3:1, with stated exceptions.

Use readable type sizes, sufficient spacing, sensible line lengths and a layout that reflows on narrow screens.

Colour should not be the sole signal of status; pair it with text, icons or other explicit information.

KEY TAKEAWAY • A design is not readable merely because it looks attractive on one monitor.

Alternative text and complex media

Meaningful images need concise alternative text that communicates the same essential purpose as the image in its context.

Decorative graphics should not produce noisy descriptions; complex archival charts may need longer adjacent descriptions.

Video needs appropriate captions, transcripts and, where needed, description of important visual-only information.

KEY TAKEAWAY • Provide equivalent information for users who cannot use the original media channel.

Ethics, copyright and rights notices

An author can only use media they created, received permission to use, or can otherwise lawfully include under applicable rules.

Archive collections may require contextual warnings, privacy screening and explicit reproduction or access restrictions.

Record source, rights basis and attribution; an interactive interface does not automatically clear copyright.

KEY TAKEAWAY • Rights and privacy screening are release gates, not optional captions.

Metadata, provenance and content integrity

For each asset record title/identifier, creator or source, date/uncertainty, rights, format, description and its screen relationship.

Use stable identifiers and readable filenames so linked assets can be updated without losing track of context.

Do not present a transformed image or shortened recording as an untouched original; label interpretations and derivatives.

KEY TAKEAWAY • A well-authored system lets users understand where content came from.

Testing navigation with a graph

List every screen (node) and every intended link (directed edge); verify that each destination actually exists.

Test reachability from Home: if an item can only be opened using a hidden URL, it is effectively an orphan for normal users.

Cycles are not automatically errors—Home → Gallery → Home is useful; dead ends and unclear loops are problems.

KEY TAKEAWAY • A sitemap is also a systematic link-testing tool.

Worked example: five-screen library orientation

Screens: Home, Search, Borrow, Help and Credits. Home offers Search/Borrow; every task screen offers Help and Home.

User task: start at Home, learn a catalogue search step, reach Help, return Home. Capture whether actions are predictable.

Include a short captioned or textual tutorial, clear labels and an attribution block for any media.

KEY TAKEAWAY • A small tested guide is better than an ambitious broken prototype.

Worked example: five-screen archival showcase

Screens: Home, Periods, Item, Context and Rights. The item page links to Context and Rights and back to Periods.

Display a fictional photograph with title, approximate date, description, and a flag that it is a teaching simulation.

Test that restrictions are visible before access and that context is not separated from the displayed item.

KEY TAKEAWAY • Hypermedia should deepen historical understanding, not just produce clicks.

Evaluate an interactive prototype

Recruit a peer to complete three tasks without assistance; note task success, time/steps and any hesitation or incorrect paths.

Then test keyboard-only navigation, readable zoom, descriptive labels, mobile layout, contrast and available alternatives.

Record defects with severity, reproduction steps, fixes and a re-test result; avoid claiming full accessibility certification.

KEY TAKEAWAY • Document evidence of user success and repairs.

A practical authoring workflow

1. Specify users and goals → 2. Audit approved assets → 3. Draw sitemap and user journey → 4. Storyboard screens.

5. Author interface and link assets → 6. Check rights and metadata → 7. Test task flow and accessibility → 8. Publish an approved copy.

Archive the source project, final web package, manifest, feedback and change history separately as needed.

KEY TAKEAWAY • Make the production process repeatable and reviewable.

Simple HTML link and button examples

Navigation: Open library guide creates a destination link with meaningful anchor text.

Action: <button type="button">Show item details</button> changes content when wired to a real event handler.

Do not make a clickable without keyboard semantics. Ensure buttons operate correctly and dynamic feedback is announced.

KEY TAKEAWAY • Use the right built-in HTML element before adding complicated scripting.

Plan a low-bandwidth and offline experience

Keep essential descriptions in text, resize access images appropriately and avoid unnecessary autoplay or huge background video.

Include captions/transcripts and clear downloads so users can choose a lower-data path.

Prefer relative links for offline and GitHub Pages portability; test after moving the folder and on a phone.

KEY TAKEAWAY • Reliable delivery is part of usable multimedia authoring.

Guided class practical

Build a five-screen prototype using fictional or cleared assets, a simple sitemap, logical labels and working navigation.

Use the supplied planning template, test the complete path from Home to a relevant item, and report any broken/orphan link.

Have a peer attempt one BLISS/BRIM task keyboard-only; record failures and a specific improvement.

KEY TAKEAWAY • Evidence matters: submit the working prototype and a concise test report.

Professional assignment: choose a path

BLISS: design a navigable digital library orientation with searchable resource guidance and help information.

BRIM: design a fictional archives mini-exhibition with contextual item descriptions, rights statement and source-to-item links.

Both must include a sitemap, storyboard, accessible controls, a testing log, rights/attribution and a short reflection.

KEY TAKEAWAY • Different scenarios, shared assessment expectations.

Common errors and their corrections

- Broken navigation: give every button an action and every link a valid destination; inspect path and filename case.
- Inaccessible controls: test keyboard focus and activation, labelled fields, image alternatives and appropriate contrast.
- Misleading content: display context, rights and restrictions rather than placing them in an undiscoverable footer.

KEY TAKEAWAY • Revise for functionality, clarity and trust before adding more effects.

Key concepts and professional practice

Authoring links text, images, sound and video into a structured, interactive resource that serves user goals.

A good prototype combines information architecture, meaningful navigation, consistent design, accessibility and ethical media use.

Reliable captured media is an important foundation for interactive guides and exhibitions.

KEY TAKEAWAY • Author for real users; verify navigation and make content understandable.